NETWORK RECONNAISSANCE & SECURITY ASSESSMENT

Project: Network Reconnaissance Using Nmap
Assessment Date: 8 November 2026
Tool: Nmap 7.993
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1. Project Overview

This project involved conducting a basic network reconnaissance and security assessment of a personal iPhone XR hotspot using Nmap. The objective was to identify active devices, discover open ports, detect network services and investigate potential security concerns.

2. Scope and Methodology

Assessment environment:

- **Target: Apple iPhone XR (personal hotspot gateway)**
- **Gateway IP: 172.20.10.1**
- **Scanning device: Lenovo ThinkPad X1 Carbon**
- **Operating system: Windows 11 Pro**
- **Network range: 172.20.10.0/28**
- **Tool: Nmap 7.993**

The assessment involved network discovery, TCP and UDP scanning, service detection, FTP investigation and DNS analysis. Testing was limited to authorised reconnaissance; no exploitation or modification of services was performed.

Key commands used:

- **nmap -sn 172.20.10.0/28 — Network discovery**
- **nmap -Pn -sV ... — Service and version detection**
- **nmap -Pn -sV -sC ... — Default script investigation**
- **nmap -Pn -sU -p 53 ... — UDP DNS investigation**
- **nmap -oN ... — Saving scan results**

3. Network Discovery Results

The final discovery scan identified two active hosts:

IP Address	Device / Observation
172.20.10.1	Likely iPhone hotspot gateway
172.20.10.5	Scanning laptop

A previous scan also detected 172.20.10.2, which did not respond during the later scan.

Final scan summary: 16 IP addresses scanned, 2 active hosts.

4. Port and Service Analysis

The gateway scan identified the following open ports:

Port	Protocol	Service / Result
21	TCP	FTP suspected, unconfirmed
53	TCP	DNS response received
53	UDP	DNS open
49152	TCP	tcpwrapped; unidentified
62078	TCP	tcpwrapped; unidentified

Additional investigations were performed:

- FTP anonymous-access and system-information checks did not return identifiable details.
- DNS returned an NXDOMAIN response during TCP investigation.
- Ports 49152 and 62078 remained unidentified.

5. Security Findings and Risk Assessment

Finding	Observation	Assessment
F001	Suspected FTP service on port 21	Undetermined
F002	DNS accessible over TCP and UDP	Informational
F003	Unidentified service on port 49152	Undetermined
F004	Unidentified service on port 62078	Undetermined

F005 Different host discovery results across scans Informational

No confirmed vulnerabilities were established during the assessment.

6. Recommendations

1. **Investigate the suspected FTP service and verify whether secure authentication and encryption are used.**
2. **Maintain updated device software and security settings.**
3. **Use a strong, unique personal hotspot password.**
4. **Investigate unidentified open ports to establish their purpose.**
5. **Conduct periodic authorised network scans to monitor changes.**
6. **Perform further validation before classifying any observation as a vulnerability.**

7. Limitations

The assessment was limited to a personal hotspot network and Nmap-based reconnaissance. The underlying services on some ports could not be identified, and no authenticated configuration review or exploitation was performed. The results therefore do not constitute a comprehensive security audit.

8. Conclusion

The project successfully demonstrated practical use of Nmap for network discovery, port scanning, service detection and preliminary security assessment.

The exercise identified active network hosts, exposed services and areas requiring further investigation. It provided hands-on experience in network reconnaissance, security analysis, risk documentation and technical reporting.

9. Evidence and Tools

Evidence files:

- **host-discovery.txt**
- **hotspot-gateway-services.txt**

Tools: Nmap 7.993, Windows Command Prompt and Notepad.